

FACTIBILIDAD

CAP 3

FORMA ESTÁNDAR

$$\text{MAX } Z = Cx$$

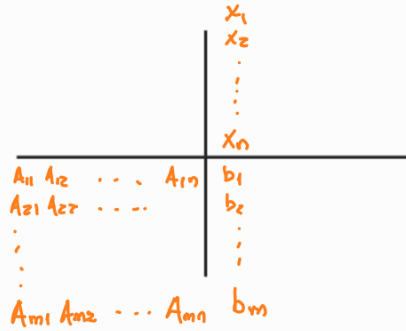
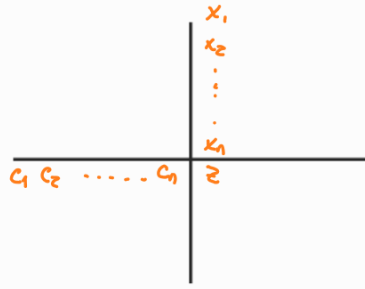
VEC VEC VEC

$$\text{s.t. } Ax = b$$

MAT VEC VEC

$$x \geq 0$$

$A_{m \times n}$



* CAP 3

A_i columnas de B (i)

A_j columnas de N(j)

$$A_{m \times n} = \left[\begin{array}{c|c} B & N \end{array} \right]$$

$m \times m$ $m \times (n-m)$

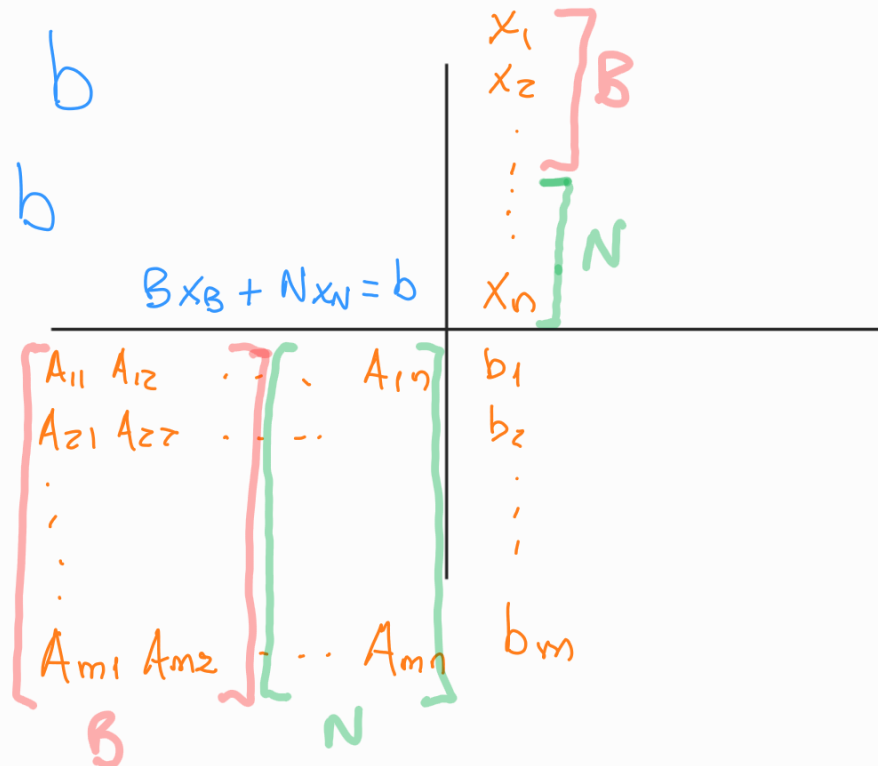
m $n-m$

n

m

$$Ax = b$$

$$Bx_B + Nx_N = b$$



$$AX = b \quad (CAP \geq 7)$$

$$Bx_B + \underbrace{N x_N}_{x_N = 0} = b$$

$$Bx_B = b \quad (2)$$

$$N_j = B Y_j \quad (1) \rightarrow N = B Y$$

$$B^{-1} N_j = \underbrace{B^{-1} B}_I Y_j$$

$$B^{-1} N_j = Y_j$$

$$Y_j = B^{-1} N_j$$

	Y_{11}	Y_{12}	$\dots$	$Y_1(n-m)$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
Y_{m1}	$\dots$	$Y_m(n-m)$				
B_{11}	B_{12}	$\dots$	B_{1m}	N_{11}	$\dots$	$N_1(n-m)$
B_{21}	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
B_{m1}	$\dots$	B_{mm}	N_{m1}	$\dots$	$N_m(n-m)$	

N_j

$$B Y_j = N_j \quad (1)$$

$$B x_B = b \quad (2)$$

θ_j escalar

$$\theta_j \geq 0$$

esté asociado a la
columna j de la NO BASE

$$\theta_j \leftrightarrow N_j$$

$$(2) - (1) \cdot \theta_j$$

$$B x_B - B Y_j \theta_j = b - N_j \theta_j$$

$$B (x_B - Y_j \theta_j) = b - N_j \theta_j$$

$$B (x_B - Y_j \theta_j) + N_j \theta_j = b$$

MAT VEC VEC ESC VEC ESC VEC

$$\begin{array}{l} x_1 - Y_{1j} \theta_j \\ x_2 - Y_{2j} \theta_j \\ \vdots \end{array}$$

$$\begin{array}{c|c}
 & X_m - Y_{mj} \theta_j \\
 \hline
 B_{m1} \ B_{12} \ \dots \ B_{1m} & \\
 \vdots & \\
 B_{m1} \ \dots \ B_{mm} &
 \end{array}$$

$$B(X_B - Y_j \theta_j) + N_j \theta_j = b$$

$$B_1(X_1 - Y_{1j} \theta_j) + B_2(X_2 - Y_{2j} \theta_j) + \dots + B_r(X_r - Y_{rj} \theta_j) \dots + B_m(X_m - Y_{mj} \theta_j) + N_j \theta_j = b$$

CONDICIÓN DE FACTIBILIDAD $\theta_j = \min \left\{ \frac{x_i}{Y_{ij}}, Y_{ij} > 0, i=1 \dots m \right\}$

Supongamos que θ_j que cumple es el de r.

$$\Rightarrow \frac{x_r}{Y_{rj}}$$

$$B_1(X_1 - Y_{1j} \frac{x_r}{Y_{rj}}) + B_2(X_2 - Y_{2j} \frac{x_r}{Y_{rj}}) + \dots + B_r(X_r - Y_{rj} \frac{x_r}{Y_{rj}}) \dots + B_m(X_m - Y_{mj} \frac{x_r}{Y_{rj}}) + N_j \frac{x_r}{Y_{rj}} = b$$

$\underbrace{\hspace{10em}}_{=0}$

$$\sum_i B_i (X_i - Y_{ij} \theta_j) + N_j \theta_j = b$$

si $\theta_j = 0 \Rightarrow \sum_i B_i X_i = b = B X_B = b$

Problema original

Max $z = Cx \longrightarrow z = C_B X_B + C_N \underbrace{X_N}_{=0}$
 s. a $Ax = b$
 $b \geq 0$
 $x \geq 0$

$$\begin{array}{c|c}
 x_1 \\
 x_2 \\
 \vdots \\
 x_n \\
 \hline
 c_1 \ c_2 \ \dots \ c_n \quad z
 \end{array}$$

$z = C_B X_B$
 $= \sum_i [c_i x_i]$

(i: base
j: no base)

$$\rightarrow \sum_i [c_i (X_i - Y_{ij} \theta_j)] + C_j \theta_j$$

$$\sum_i [c_i X_i - c_i Y_{ij} \theta_j] + C_j \theta_j$$

$z_{original} \rightarrow z_0$

$z_{nuevo} \rightarrow z$

$$\sum_i c_i x_i - \theta_j \sum_i c_i y_{ij} + c_j \theta_j$$

$$\sum_i c_i x_i + \theta_j \left(c_j - \underbrace{\sum_i c_i y_{ij}}_{z_j} \right)$$

$$\sum_i c_i x_i + \theta_j (c_j - z_j)$$

$$z = \underset{\downarrow}{z_0} + \theta_j (c_j - z_j)$$

$$z = z_0 + \theta_j (c_j - z_j)$$

CONDICIÓN DE
OPTIMIZACIÓN

